

STUDY ON THE DIVERGENCE IN POPULATION ESTIMATES ARISING FROM THE CENSUS AND NSS SURVEY

A PROJECT

Submitted by

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Abstract

The accurate estimation of a nation's population is fundamental for effective governance, policy formulation, and resource allocation. In India, population data is primarily derived from the Census of India, which provides a complete count every ten years, and from the National Sample Survey (NSS) conducted by the National Statistical Office (NSO), which offers demographic and socio-economic estimates based on sample data rather than actual population counts. The Census provides complete population counts, while the NSS offers demographic insights based on sample surveys. However, population estimates from the NSS may differ from the Census due to sampling and methodological differences. This report aims to identify and analyse the various factors contributing to these discrepancies, assess their implications for policy and planning, and propose recommendations to improve the alignment of population data from these two key sources.

CHAPTER - I

INTRODUCTION

In India, the census is the primary source of official population counts. The Census of India stands as the most extensive source of information on the nation's population, conducted every ten years since 1872. Since independence, the census has been conducted by the Register General and Census Commissioner of India (RGI) under the Ministry of Home Affairs. This complete enumeration provides a wealth of demographic and socio-economic data, reaching down to the village level and offering a comprehensive snapshot of the country's inhabitants.

India estimates and updates population figures using methods like the Sample Registration System (SRS), Civil Registration System (CRS), demographic surveys (e.g., NSS), and administrative records. Complementing this effort is the National Sample Survey (NSS), undertaken by the National Statistical Office (NSO), which conducts large-scale sample surveys on a variety of socio-economic indicators, often with annual or quinquennial periodicity. Both the Census and the NSS play indispensable roles in informing planning processes, guiding policymaking, and evaluating the progress of developmental initiatives across the country.

Accurate population estimates are extremely important for many reasons. These estimates are essential for shaping policy formulation across various sectors, enabling governments to allocate resources effectively based on the needs of different regions and demographic groups. Furthermore, population data are very important for setting the boundaries of voting areas, making sure that people are fairly represented. They also help decide who gets benefits from different government welfare schemes meant to support poor and vulnerable groups. In addition to these direct uses, Census data act as a key reference point for many

other surveys and studies. They give a basic picture of the population, which helps in understanding and comparing other data.

Despite the critical roles played by both the Census and the NSS, observations have revealed divergences in their population estimates. NSS estimates are often lower than the figures from the Census or the population projections made using Census data. These differences are not just in the total population figures but also in specific areas, such as the estimated number of people living in slums and the reported rates of disability, raising concerns about the accuracy of the data. Understanding the reasons behind these divergences is essential for ensuring the reliability of the data used for crucial decision-making processes.

This study is therefore undertaken with the primary objective of identifying and analysing the various factors that contribute to the divergence in population estimates arising from the Census and the NSS survey. Besides identifying the causes of these differences, the report will also examine how they affect policy-making and planning in various sectors. Ultimately, the study aims to provide well-reasoned and actionable recommendations that can be implemented to improve the alignment of population estimates derived from these two key statistical sources in India.

Objective of the study:

To examine the reasons behind the divergence in population estimates between the Census of India and the National Sample Survey (NSS), even though the NSS does not provide official population counts, and to analyse how methodological differences, sampling practices, and definitional issues contribute to this discrepancy. The study also seeks to evaluate how these differences affect the use of data in policymaking and to propose recommendations for better alignment between the two data sources.

Overview:**Census:**

The census is a complete enumeration of a population in India. The first official, complete census of India was conducted in 1881. The census was conducted once every ten years until 2011. Census, an enumeration of people, houses, firms, or other important items in a country at a particular time. The data collected through the Census is used for administration, governance, planning and policy-making as well as management and evaluation of various programmes run or to be introduced by the Government, NGOs, researchers, commercial and private enterprises, etc.

NSS survey:

The national sample survey is conducted by Ministry of statistics and programme implementation. The National Sample Survey Organisation (NSSO) has been conducting nation-wide sample surveys on various socio-economic aspects since 1950. These surveys are conducted in the form of rounds extending normally over a period of one year though in certain cases the survey period was six months.

Key differences between Census and NSS survey:

The census collects data from every member of the population and NSS collects data from portion of the population. The census conducts once every ten year and NSS conducts regular survey on a variety of topics throughout the year. The census collects demographic characteristics and NSS survey provides specific estimates and analysis on various socio-economic issues.

Why Accurate Population Estimates Are Important?

In India, the Census and the National Sample Survey (NSS) are two key sources of demographic and socio-economic information. The Census provides a

complete enumeration every ten years, while the NSS offers more frequent, detailed data on various aspects of household and individual conditions through carefully designed sample surveys. While they serve different purposes, maintaining alignment between the two sources is critical for the following reasons:

Supporting Policy Consistency and Accuracy:

Government policies and programs are designed using data from both the Census and NSS. Variations between the two can lead to inconsistencies in setting development goals, especially in areas like health, education, employment, and social protection.

Ensuring Effective Delivery of Welfare Schemes:

Planning and monitoring of welfare schemes often draw on Census data for broad coverage and NSS data for tracking patterns and progress. Discrepancies in estimates can result in challenges in identifying beneficiaries and delivering services efficiently.

Enabling Fair Allocation of Resources:

Accurate estimates help in assessing the needs of different regions. Inconsistent data between the Census and NSS may lead to over- or under-allocation of essential resources like schools, healthcare facilities, and food distribution systems.

Informing Sampling Design and Estimation Methods:

Sample surveys rely on demographic distributions to design weights and draw meaningful conclusions. If the reference data used for this purpose differ substantially from actual conditions, it can influence the reliability of survey-based indicators such as employment or consumption trends.

Facilitating Planning at Local Levels:

While the Census provides detailed data at the village or ward level, the NSS is frequently used to understand patterns in living conditions across different regions and population groups. A mismatch between the two sources can affect local-level planning and the accuracy of interventions.

Strengthening Public Confidence in Data:

When different official sources show large gaps in key estimates, it can raise concerns about data quality and reduce trust in public statistics. This can also affect participation in future data collection efforts.

Monitoring Progress Toward National Goals:

Tracking India's development goals and international commitments such as the Sustainable Development Goals (SDGs) depends on reliable and comparable data. Alignment between NSS and Census estimates helps maintain the integrity of these measurements and guides timely policy action.

Accurate and consistent population-related estimates are vital for data-driven decision-making, fair distribution of public goods, and effective governance. While the Census and NSS serve distinct yet complementary roles, minimizing divergence between them enhances the overall strength of India's statistical system and ensures better planning and delivery of services.

CHAPTER – II

LITERATURE REVIEW

Several studies have documented notable discrepancies between population-related estimates derived from the Census of India and the National Sample Survey (NSS). While the Census provides a complete enumeration of the population, the NSS collects data from sample households, and although it does not provide population totals, it uses population projections (typically based on the most recent Census) for weighting and estimation.

1. Discrepancies in Estimates:

A consistent finding in the literature is that NSS-based estimates tend to be lower than those derived from the Census. These differences are observed not just in total population-related projections but also in subgroups such as slum populations, Scheduled Castes, rural populations, and disabled persons (Asher et al., 2019; Kumar & Sharma, 2024). This underestimation varies across regions and demographic segments.

2. Methodological Differences:

A key source of divergence stems from methodological differences. The Census aims for complete coverage with a de facto approach, while the NSS uses stratified multi-stage sampling with limitations in coverage, especially in remote or rapidly changing areas. The sampling frame of NSS, especially in urban areas (Urban Frame Survey blocks), is often not updated in real-time, missing newly developed or reclassified regions (IHD, 2023; MoSPI, 2021).

3. Definitional Issues:

Studies point to inconsistent definitions of key terms like “household,” “urban area,” “worker,” and “slum” across the two systems (Teitelbaum, 2022). These definitional mismatches result in inconsistencies in identifying and

categorizing population groups, especially in complex urban settings or among mobile populations.

4. Coverage and non-response:

The NSS excludes institutional populations (e.g., in prisons), which are included in the Census. Additionally, non-response rates have been reported to be rising in NSS, especially in urban and high-income areas, affecting the accuracy of estimates (MOSPI, 2023). Meanwhile, the Census too faces challenges in enumeration, especially in remote and marginalized communities.

5. Use of Census Data in NSS:

Though the NSS does not estimate population size, it depends heavily on Census-derived projections for its estimation weights. If these Census figures are outdated (as is currently the case with the 2011 data still being used), it may result in misalignment of NSS estimates with current realities (India Forum, 2022).

6. Sampling and Estimation Techniques:

Techniques such as Probability Proportional to Size (PPS) used by NSS can lead to biases if the size data are outdated. Differences in estimation procedures, such as the use of post-stratification weights and the treatment of non-response, also contribute to the divergence in estimates (UN Stats, 2019).

CHAPTER – III

METHODOLOGY

This study adopts a descriptive and analytical approach to explore the reasons behind the differences in population figures reported by two major sources in India the Census of India and the National Sample Survey (NSS). The research is based entirely on secondary data, making use of information that has already been collected and published. Specifically, we will draw from official sources such as the Registrar General of India (RGI) and the National Statistical Office (NSO). These include official reports, survey manuals, operational guidelines, technical documents, and relevant academic literature. The primary focus will be on understanding how population estimates are derived in both the Census and the NSS, including a detailed comparison of their methodologies, sampling strategies, and data collection techniques.

A central component of this study involves a critical review of the design and implementation of both the Census and NSS. We will examine the official definitions used to identify and count populations, the procedures for drawing samples in the case of surveys, and the enumeration practices followed in full counts like the Census. By dissecting these elements, we aim to uncover structural and procedural differences that may contribute to the observed disparities in population data.

Our research will methodically analyse the foundational documents that guide both the Census and NSS, such as instructions to enumerators, methodological handbooks, and sample design frameworks. This careful investigation will allow us to pinpoint inconsistencies and gaps in the data collection processes of both systems. In doing so, we will also consider the timelines, coverage, periodicity, and resource allocation associated with each

system, as these factors can significantly influence the accuracy and reliability of the resulting data.

Beyond simply identifying the technical reasons for the discrepancies, our study seeks to interpret their broader implications. We will assess how differing population numbers impact government planning, policy formulation, and the allocation of public resources. Accurate and consistent population data are crucial for effective governance, especially in areas such as healthcare, education, employment, housing, and social welfare.

Ultimately, this research aims to contribute to a deeper understanding of the divergence in population estimates between the Census and NSS. By shedding light on the causes and consequences of these discrepancies, we intend to propose practical recommendations for improving the consistency, comparability, and reliability of population data in India. Such improvements are vital to ensuring that policies based on these data are well-informed and equitable.

CHAPTER – IV

ANALYSIS AND DISCUSSION

Analysis of Divergence arising from Census and NSS survey:

Hamlet group/sub division formation and population estimation:

In rural India, villages often consist of multiple scattered hamlets, and this unique settlement pattern poses a major challenge for accurate population estimation. These hamlets, which can range from a few households to substantial clusters, may be geographically dispersed and lack clearly defined boundaries, complicating the task of enumeration. This characteristic of rural habitation can contribute to the divergence in population estimates between the Census and the NSS in several ways.

In the Census, which aims for a complete count, the scattered nature of hamlets increases the potential for undercounting if enumerators unintentionally miss remote or less accessible settlements. The large number of habitations spread across India's diverse landscape makes it challenging to ensure that every household is included. If some hamlets are missed during the Census, the total population count will naturally be lower than the real number.

The NSS, on the other hand, employs a methodology that involves hamlet-group formation and selection in villages identified as large (typically with a population of 1200 or more). According to NSSO guidelines, such large villages are divided into a suitable number of hamlet-groups, and then a sample of two hamlet-groups is selected for the survey. While this sampling approach is designed to manage workload and ensure representation, it carries the risk of not fully capturing the population variability within a large village. When the selected hamlet-groups properly represent the village in terms of population size and

composition, the estimates for the whole village can closely reflect what would be found through a complete enumeration.

The hamlet group/sub division formation done only in rural areas. The present population is collected from village VAO or Anganwadi. These details not always true. After we do the hamlet group/sub division formation in the village with the present population and two hamlets selected randomly. After we do the listing and here, we got the problem that undercount population. So, this makes the divergence in the population estimation. We need to get the approximate present population in the village. From getting the approximate population, we can reduce the divergence in the population estimation. The NSS surveys estimate population figures by collecting data from a representative sample of villages and towns across the country.

The decentralized and often informal growth of rural settlements in India creates major challenges for both the complete coverage required by the Census and the representative sampling targeted by the NSS, which can lead to coverage errors and differences in their population estimates. However, by ensuring more accurate and up-to-date information on the present population during hamlet-group formation, and improving the representativeness of selected hamlets, these differences can be reduced.

Outdated sampling frames:

The National Sample Survey (NSS) relies on the Urban Frame Survey (UFS) blocks as the primary sampling frame for conducting surveys in urban areas across India. The UFS is a dedicated survey undertaken by the NSSO in five-year phases with the aim of preparing and maintaining a comprehensive frame of compact urban geographical units. These units, known as UFS blocks, serve as the first-stage sampling units for various socio-economic surveys conducted by the NSS. While the UFS plays a vital role in providing a structured

and systematic framework for urban sampling, its periodic (rather than continuous) update cycle can pose certain limitations, especially in light of India's rapidly changing urban landscape.

One challenge associated with the UFS update cycle is its limited ability to capture newly urbanized areas in real time, particularly the 'Census Towns' identified by the Census of India. Between two Census rounds, substantial urbanization often occurs, and many villages may meet the criteria to be classified as Census Towns. Since the UFS is typically revised every five years, there may be a lag in incorporating these newly emerging urban areas into the sampling frame. As a result, some areas might still be treated as rural in the NSS framework until the next UFS update, potentially leading to partial coverage of the urban population in NSS estimates.

The period between Census exercises often witnesses significant urbanization, with numerous villages meeting these criteria and being classified as Census Towns in the latest Census. However, the UFS frame, which is typically updated in five-year cycles, might not promptly incorporate these newly urbanized areas. This time lag in updating the UFS to include these new urban entities can result in an under coverage of the urban population in NSS estimates, as these areas might still be treated as rural in the NSS sampling frame until the next UFS update.

Additionally, even within long-established urban areas, the UFS used by the NSS may rely on older phases (such as the 2012–17 phase), despite the availability of more recent data. Given the rapid development and demographic shifts occurring in urban India, relying on older UFS frames can reduce the accuracy of urban representation in the NSS samples and may contribute to underestimation of the current urban population.

However, it is important to note that the UFS provides a well-organized and standardized sampling structure that enables consistent data collection across urban India. With timely updates and better integration of newly urbanized areas, the UFS-based approach of the NSS holds significant potential for capturing evolving urban dynamics more accurately in future survey rounds.

Sampling Design:

A fundamental distinction between the Census of India and the National Sample Survey (NSS) lies in sampling design. The census provides a complete and detailed account of the population at a specific point in time.

In contrast, the NSS employs a stratified multi-stage sampling design to select a representative subset of households from the population. This involves dividing the population into strata based on certain characteristics and then selecting samples in multiple stages to ensure representation of different segments of the population. Although the NSS sampling design aims to produce estimates that represent the overall population, the use of a sample naturally introduces some degree of sampling variability. This means the population estimate from an NSS survey is just one of many possible outcomes that could result from different samples drawn using the same method. Because of this sampling variability, differences may naturally arise when these estimates are compared to the Census, which aims to count the entire population.

The NSS uses Probability Proportional to Size (PPS) sampling to select primary units' villages in rural areas and UFS blocks in urban areas. In this method, units with larger populations or more households (based on the latest Census or UFS) have a higher chance of being selected, helping to improve the survey's representativeness and efficiency. Although PPS sampling is widely used and effective, it can sometimes introduce bias. If the size data used to select areas (like population from the last Census) is outdated, it may not reflect current

conditions due to changes like growth or migration. This can cause some areas or groups to be over- or underrepresented in the NSS sample, leading to differences from Census estimates. The effectiveness of PPS sampling depends on how well the size measure (like population) relates to the characteristics being studied. If there's a weak link for example, a large village may not have many households of a target group PPS may introduce bias instead of reducing it. NSS also uses PPS with replacement, meaning the same area can be selected more than once. This can lead to bias if such areas aren't representative of similar-sized ones. To correct this, NSSO uses methods like the Horvitz-Thompson estimator to adjust for unequal selection chances. Still, these estimates rely heavily on the quality of the size data used, which, if outdated, can affect the accuracy and cause differences from Census results.

Therefore, the fundamental difference in approach between a complete enumeration in the Census and a sample survey in the NSS, along with the specific design elements of the NSS sampling methodology, are key factors that can lead to a divergence in their population estimates.

Sample size:

The National Sample Survey (NSS) aims to provide reliable estimates of various socio-economic indicators at the national and state levels. However, it faces constraints such as limited time, budget, and manpower for fieldwork and data processing. Due to these limitations, the survey generally uses sample sizes that are large enough to provide reliable results at broader levels, but may have some limitations when estimating for smaller areas or specific population groups.

In statistics, it's well known that sample size affects how accurate survey results are. Larger samples usually give more precise estimates, with smaller standard errors. This means we can be more confident that the survey result is close to the actual population value. On the other hand, smaller samples often

result in estimates with larger standard errors, which can make it more challenging to determine how closely the results reflect the true population value.

When looking at smaller areas or specific groups, the limited sample size can cause greater variation in NSS estimates, making them more variable. The limited sample size in the NSS, mainly due to practical constraints such as time, money, and staff, helps explain why population estimates may vary for smaller areas or specific groups. However, despite these challenges, the NSS methodology is designed to provide reliable and useful estimates for broader population trends.

Sampling and non-sampling error:

The Population estimates from the National Sample Survey (NSS) are naturally affected by sampling errors. This is a direct consequence of the fact that the NSS is based on data collected from a representative sample of the population rather than a complete enumeration, as is the case with the Census of India. Sampling error reflects the natural variation that occurs when a subset of the population is used to make inferences about the entire group. Due to this variability, the estimates obtained from an NSS sample might differ to some extent from the true population values that are captured by the Census. The magnitude of sampling error is influenced by factors such as the sample size, the sampling design, and the variability within the population being studied. Larger sample sizes generally lead to smaller sampling errors, and stratified sampling techniques can help to reduce this error by ensuring representation across different subgroups.

In addition to sampling errors, both the Census and the NSS are affected by non-sampling errors, which can occur at any stage of the data collection, processing, and analysis. Unlike sampling errors, which can be mathematically estimated, non-sampling errors are often more challenging to quantify.

The National Statistical Office (NSO), which conducts the NSS, acknowledges the potential for under-coverage in its population totals. This under-coverage can stem from several factors, including imperfections in the sampling frame used, under-enumeration of households within a selected sample first-stage unit, and under-enumeration of members within a sampled household. These issues contribute to the overall non-sampling error in the NSS estimates. Similarly, the Census of India, though a complete count, is also subject to non-sampling errors. These can result from a shortage of well-trained enumerators, low response rates, widespread illiteracy, and the difficulty of covering a widely dispersed population with limited resources.

The divergence in population estimates between the Census and the NSS can be partly attributed to the combined effect of sampling errors in the NSS and the differing magnitudes and types of non-sampling errors that affect both data sources. For example, if the Census has a higher undercount rate for certain demographic groups than the NSS, or if the NSS faces greater non-response bias in a particular segment of the population, the population estimates for these groups and therefore the overall totals are likely to differ. However, both the Census and the NSS follow systematic methodologies and implement quality checks to reduce such errors, and when considered within their respective frameworks and objectives, both sources offer meaningful and complementary information about the population.

Estimation and weighting techniques:

The National Sample Survey (NSS), being a sample-based inquiry, relies on various estimation and weighting techniques to extrapolate the data collected from the sampled households to represent the entire population. These techniques involve the application of survey weights to each sampled unit. These weights are typically derived based on the survey's sampling design, taking into account the probability of selection at each stage. Additionally, weighting procedures often

incorporate auxiliary information, which frequently includes population data from the most recent Census, to adjust the sample distribution to match known population characteristics.

The choice of variables used for weighting and the specific methodology employed to adjust for non-response are critical aspects of the estimation process that can significantly impact the final population estimates derived from the NSS. If the Census data used to guide the weighting of the NSS sample are outdated because the Census is conducted only once every ten years or if those figures contain errors, these issues can be carried over into the NSS estimates. For instance, if the urban population has grown substantially since the last Census, using older Census figures for weighting might lead to an underestimation of the current urban population in the NSS.

Moreover, the application of survey weights might vary across different rounds of the NSS or across different types of surveys conducted by the NSSO. Differences in estimation and weighting methods across NSS surveys can lead to inconsistencies in population figures, which may contribute to the overall gap when compared with Census data. However, these weighting procedures are carefully designed to adjust for sampling design and non-response, and when applied appropriately, they help improve the accuracy and representativeness of the survey estimates.

Differences in reference period:

A notable methodological difference between the Census of India and the National Sample Survey (NSS) lies in the reference period used for population enumeration. The Census typically adopts a specific reference date, aiming to capture the population count at a particular point in time. For instance, the reference date for the Census 2011 was March 1, 2011. This approach provides a summary of the population on that specific date.

In contrast, the NSS employs varying reference periods for different types of data collected in its surveys. For example, in employment surveys, the reference period might be 'last week' or 'last year' preceding the date of the survey.

These differences in reference periods can contribute to the divergence in population estimates, particularly when considering temporary populations such as migrants or seasonal workers. The Census, with its single reference date, will count individuals based on their presence at their usual place of residence on that specific day. However, the NSS, with its longer and varied reference periods, might capture these individuals based on their status or movement over a more extended duration. Therefore, the basic difference in the timing of data collection between the Census and the NSS is a key reason for the variations in their population estimates.

Problems in Definitions:

A fundamental source of divergence in population estimates between the Census of India and the National Sample Survey (NSS) arises from differences in the definitions of key terms used by the two agencies. The lack of complete harmonization in these definitions can lead to variations in how geographical areas, households, and individuals are classified and enumerated, ultimately impacting the comparability of their population statistics.

For example, one significant area of definitional difference lies in the concept of a 'household'. In the Census of India, a household is defined as a group of persons who normally live together and take their meals from a common kitchen, unless exigencies of work prevent any of them from doing so. This definition emphasizes common residence and shared meals as key identifiers of household membership.

In contrast, the National Sample Survey (NSS) defines a household as a group of persons normally living together and sharing a common kitchen, but

with added emphasis on economic decision-making and budget-sharing. For instance, in NSS surveys, domestic servants, paying guests, or unrelated individuals living in the same premises but managing their own food and finances are treated as separate households, even if they share the same dwelling unit. This difference in classification can lead to variations in household counts and associated population estimates, especially in densely populated urban areas where multiple families or individuals may co-reside in shared spaces. Such definitional nuances contribute to divergence in statistics between the two data sources.

In conducting urban surveys, the NSS uses Urban Frame Survey blocks as its sampling frame. While these blocks are generally aligned with the urban areas identified by the Census, the timing and procedures of updating the Urban Frame Survey may not always capture newly emerging urban settlements promptly. This can sometimes lead to differences in urban and rural classification between the two sources. Additionally, variations in how concepts like household and family are defined, such as how individuals living together but managing their own cooking or finances are counted, can also cause differences in population estimates.

Another area where definitions differ is in the identification of slums. The NSS and the Census apply different criteria for classifying slums, reflecting their distinct survey goals and methods. These differences can lead to varying estimates of the population living in slum conditions. However, these definitional differences are not shortcomings but rather reflect the unique approaches of each system. Understanding these differences helps in interpreting and using the data from both the Census and NSS more effectively.

Listing variability:

A key first step in both the Census of India and the National Sample Survey (NSS) is listing households in the selected areas. This listing forms the basis for either full coverage in the Census or sampling in the NSS. The quality and accuracy of this process are crucial, as any lapses can affect survey coverage and lead to differences in population estimates between the two.

In the NSS, enumerators are responsible for identifying and listing all households in their assigned areas. However, the accuracy of this process can vary due to several factors such as the training and supervision of enumerators, the complexity of settlement layouts (like crowded urban slums or scattered rural hamlets), and the time and resources available for the work. If any households are missed during listing, it can result in under coverage and a lower population count. On the other hand, double-counting households can lead to overestimation.

In the National Sample Survey, the quality of household listing is strongly influenced by the training and supervision provided to its fieldworkers. Unlike short term staff, NSS employs full time fieldworkers who undergo detailed and ongoing training programs designed to equip them with the skills necessary for accurate and consistent data collection. These workers are better prepared to navigate the complexities of diverse settlement types such as densely populated urban slums, multifamily dwellings, and scattered rural hamlets. The NSS places significant emphasis on standardized procedures for household identification and listing, aiming to capture all households including those in informal or nontraditional living arrangements correctly and consistently across surveyed areas.

This structured approach in training and supervision helps reduce common data collection errors like missing households or counting the same household more than once. However, despite these efforts, operational challenges such as

limited time for fieldwork, logistical difficulties, and accessibility issues in certain regions can still impact the completeness of household listings. This means that while the NSS strives for accuracy through well trained personnel and robust procedures, some variability in the data can occur.

When comparing these processes with the Census, which primarily relies on a large number of temporarily trained personnel such as school teachers and follows different enumeration guidelines, the differences in training quality, supervision intensity, and household listing procedures can lead to inconsistencies. However, these differences in approach reflect the unique objectives and operational realities of each survey and contribute to the variations seen in population estimates between the NSS and the Census.

Omission sections of population:

Both the Census of India and the National Sample Survey (NSS), despite their best efforts, face the challenge of potential omission or undercounting of certain segments of the population. These sections often include vulnerable and marginalized groups who are inherently more difficult to enumerate due to their living circumstances or social status. The homeless population, for example, lacks a fixed address, making them particularly hard to reach through traditional household-based survey methods used by both the Census and the NSS. While the Census undertakes specific efforts to enumerate the homeless, the transient nature of this population increases the risk of undercounting.

Migrants, especially those who have recently moved, are undocumented, or are part of a floating population (e.g., seasonal workers, nomadic communities), also pose a significant challenge for complete enumeration in both surveys. Their movement patterns can make it difficult to ascertain their usual place of residence at the time of the Census reference date, and they might be

missed by NSS household surveys if they are not part of a settled household in the sampled areas.

The differences in population estimates between the Census and the NSS can grow if each survey misses hard-to-reach groups at different rates or in different ways. For instance, the NSS might have specific limitations in accessing or surveying certain marginalized communities due to social barriers, geographical remoteness, or lack of specific protocols for enumerating transient populations. If the Census, through its larger scale and more intensive enumeration efforts, is relatively more successful in counting these often-overlooked segments, it would contribute to a higher population estimate compared to the NSS. Conversely, there might be certain vulnerable populations that are better captured by the sampling methodology of the NSS in specific contexts. Therefore, the differential under coverage of these hard-to-enumerate sections of the population in the Census and the NSS is a factor that can lead to discrepancies in their overall and group-specific population estimates.

Inclusion and exclusion criteria:

Both the Census of India and the National Sample Survey (NSS) establish specific criteria for determining which individuals or households are to be included or excluded from their enumeration or survey. These inclusion and exclusion criteria are essential for defining the scope of the data collection and ensuring that the results are relevant to the intended population of study. However, differences in these criteria between the two agencies can lead to variations in who is counted as part of the population, thereby contributing to the divergence in their estimates.

The Census of India aims to enumerate all persons who are present within the geographical boundaries of the country at the time of the Census reference date, as well as those who are usual residents but might be temporarily absent.

This includes individuals living in households, as well as those residing in institutional settings such as prisons, hospitals, hostels, and other collective dwellings. The Census aims to count every person based on where they are on the day of the Census.

In contrast, the NSS typically focuses on surveying 'normal' households, individuals and the institutional population from its coverage. In the NSS a household usually means people who live together and share meals from the same kitchen. This can include those temporarily away but it may not cover everyone counted in the Census especially people living in institutions like jails. The NSS's primary objective is to collect socio-economic data from households, and its sampling frame and methodology are designed accordingly.

The differences in these inclusion and exclusion criteria mean that certain segments of the population that are counted in the Census might not be covered by the NSS. Therefore, the differing inclusion and exclusion criteria employed by these two key data sources are an important factor in understanding the observed divergence in their population estimates.

Non-response and its impact:

Non-response, which refers to the failure to obtain data from sampled units, is a challenge faced by both the Census of India and the National Sample Survey (NSS). This can occur when households or individuals refuse to participate in the survey, are unavailable at the time of contact, or are otherwise unable to provide the required information. The presence of non-response can introduce bias into the survey estimates if the characteristics of the non-respondents differ systematically from those of the respondents.

The impact of non-response on population estimates can vary between the Census and the NSS. If the non-response rates are significantly different between the two surveys, or if the segments of the population that are more likely to not

respond differ, it can lead to discrepancies in the final estimates. For instance, studies have shown that non-response rates in surveys tend to be higher among certain demographic groups, such as those from higher income strata or in urban areas. If the NSS experiences a higher rate of non-response from such groups compared to the Census, it could lead to an underrepresentation of these segments in the NSS sample and consequently affect the overall population estimates and the estimates for characteristics associated with these groups.

Reports indicate that non-response rates in NSS surveys have been increasing in recent years, particularly in urban areas. MOSPI itself has acknowledged this trend, noting a doubling of non-responsiveness to NSO surveys over a five-year period. Factors contributing to this include reluctance to participate, especially among higher income groups and in gated communities, as well as concerns about data privacy. In the recently conducted Household Consumption Expenditure Survey (HCES), non-response rates were reported to be as high as 9.8% in urban areas and 4.1% in rural areas. An analysis of this data showed a significant increase in non-responsiveness among the 'high income' category in urban areas compared to previous rounds.

Both the Census and the NSS encounter non-response issues; however, the Census benefits from a legal mandate and more rigorous follow-up procedures, potentially resulting in lower non-response rates. In contrast, the NSS is a voluntary survey, which may lead to higher or differing patterns of non-response. If the non-response biases vary between the two surveys, this can contribute to discrepancies in their population estimates. Although both surveys implement strategies such as follow-up visits and weighting adjustments to mitigate non-response effects, the effectiveness of these approaches may differ, thereby influencing the divergence in reported population figures.

Census delays and impacts:

The Census of India, traditionally conducted every ten years, is a foundational pillar of the nation's statistical system. However, the 2021 Census, which was due to be the 16th in the series, has faced significant delays and has been postponed indefinitely. This first-time delay has far-reaching impacts, particularly on other statistical exercises that rely on Census data, including the National Sample Survey (NSS).

The NSS relies heavily on Census data for sampling frames and aligning its estimates with known population totals. With the delay of the 2021 Census, the NSS continues to use outdated 2011 data, reducing the accuracy and representativeness of its estimates. This reliance on old data contributes to growing divergence between NSS figures and the actual population. Additionally, population projections based on outdated Census trends further impact planning and widen discrepancies, affecting the broader statistical system. The delay in the Census affects the whole statistical system and adds to the differences in population estimates between the Census and the NSS.

Impact of Divergence in the Population estimates:

In the context of the National Sample Survey NSS, differences in population estimates compared to other sources such as the Census can have practical implications for data interpretation, program design, and policy formulation. The NSS plays a crucial role in providing regular sample-based insights into various socioeconomic indicators which are widely used by researchers, administrators, and policymakers.

When population estimates from the NSS differ from those of the Census, it can affect how population-based indicators are calculated and interpreted. For example, employment rates, literacy levels, and health coverage rates often depend on accurate population figures. Any variation in estimated population size

may lead to differences in these indicators which can influence how issues are identified and prioritized in policy planning.

In addition, population figures are often used to determine funding allocations, set program targets, or plan infrastructure development. Discrepancies in estimates can create challenges in ensuring that resources are appropriately distributed, especially at the state, district, or village level. This may affect the delivery of essential services such as education, health, nutrition, and social welfare schemes.

Moreover, planning for smaller areas or specific population groups may be particularly affected if sample sizes are limited or if population estimates differ substantially. In such cases, divergence between the NSS and Census figures could influence how effectively policies reach intended beneficiaries.

However, it is important to note that the NSS and the Census are designed with different objectives. The NSS provides more frequent and detailed socioeconomic data through sample surveys, while the Census offers a complete count at periodic intervals. Both are valuable and serve complementary roles in the statistical system.

Aligning key definitions and improving consistency between the two data sources while respecting their distinct purposes can help enhance the reliability of estimates used in policy and planning. This can lead to more informed decision-making and better outcomes for governance and public welfare.

Discussions:

The difference in population estimates between the Census of India and the National Sample Survey (NSS) is due to several factors, including differences in methods, coverage, definitions, and data collection processes. The fundamental difference in methodology, a complete enumeration in the Census versus a sample survey in the NSS is a primary driver, leading to inherent sampling variability in

NSS estimates. This variability increases due to sample size limitations in the NSS, which can reduce the accuracy of estimates, especially for smaller areas or specific population groups.

Beyond the sampling approach, the scattered nature of rural settlements and the difficulties in identifying and enumerating hamlets can lead to coverage errors in both the Census and the NSS, though the extent of these errors may differ between the two. The use of outdated sampling frames in the NSS, particularly the Urban Frame Survey (UFS) for urban areas, can contribute to under coverage, especially in rapidly urbanizing regions and newly classified Census Towns where recent changes may not be captured accurately. The delay in the Census 2021 has worsened this issue by leaving the NSS to rely on increasingly outdated population benchmarks for its sampling and weighting procedures.

Differences in the definitions of key terms such as 'household', and 'worker' between the Census and NSS create further inconsistencies in classification and enumeration. The varying reference periods used by the NSS for different types of data can also affect the counting of mobile populations and those with fluctuating statuses. Additionally, both surveys face challenges related to non-response, with increasing rates in NSS potentially introducing bias. The use of PPS sampling in NSS, while efficient, is prone to biases if the size measures are outdated. Variability in household listing and differing inclusion/exclusion criteria for specific population groups also contribute to the observed divergence.

The divergence in population estimates has significant implications for policy and planning. As these estimates form the basis for resource allocation, welfare program targeting, and the assessment of developmental progress, discrepancies can lead to inefficiencies and inequities. For instance, dependence on outdated Census data due to delays can exclude eligible beneficiaries from welfare schemes. Similarly, inaccuracies in urban population estimates can affect the planning and provision of urban infrastructure and services. The use of

different population figures by various government agencies can also lead to inconsistencies in policy implementation and monitoring.

Bridging the gap between Census and NSS population estimates is challenging due to several reasons. First, the Census aims to count every individual while the NSS is based on a sample survey. However, such variations do not necessarily reflect flaws in either system but rather highlight the different roles they play. The Census is intended to capture a full enumeration and long-term demographic structure, whereas the NSS focuses on understanding trends and patterns in living conditions, employment, consumption, and other socio-economic variables through periodic surveys.

CHAPTER – V

RECOMMENDATIONS AND SUGGESTIONS

To make the population estimates from the Census and the NSS more consistent and reliable, the following suggestions are made:

Sources for Estimating Current Village Population for Survey Planning:

To conduct the village-level survey effectively, it is essential to have an accurate estimate of the current population for subgroup formation. Obtaining reliable population figures can be challenging, especially in the absence of recent official data. Typically, information is collected from local administrative sources such as the Village Administrative Officer (VAO) and Anganwadi workers.

To improve the accuracy of population estimates and minimize discrepancies between Census data and current estimates, the following additional data sources and cross-verification methods are utilized:

- Recent public gatherings or local events
- Church or religious group records
- Data from specific social groups within the village
- Ration card or voter ID records
- Satellite imagery and remote sensing estimates
- Local school enrolment data and inputs from school teachers
- Beneficiary lists from recent government schemes (e.g., *Kudumba Thalaivigal* initiative)

It is strongly suggested to increase the number of Anganwadi workers in larger or rapidly growing villages to ensure better local coverage. Additionally, Anganwadi workers should be encouraged or mandated to conduct population

updates at least once every three months. This regular and localized data collection would significantly improve the reliability of population estimates and enable more effective planning and targeting of survey activities and welfare schemes. Using multiple sources for triangulation enhances the reliability of the population estimates and supports more precise survey planning.

Updating sampling frames regularly:

The NSSO should implement more frequent updates of the Urban Frame Survey (UFS) to better reflect the rapid pace of urbanization and the emergence of new Census Towns. This could include using digital technology and satellite images to more quickly and accurately track changes in urban areas. Simultaneously, mechanisms for more dynamic updates of the rural sampling frame should be explored to account for villages transitioning into urban areas and the development of new rural settlements.

Harmonizing definitions across surveys:

A dedicated working group comprising representatives from the Census office and the NSSO should be established to undertake a comprehensive review and harmonization of the definitions of key terms that are critical for population estimation, such as 'urban area', 'household', 'worker', and 'slum'. Following this, a common glossary of these definitions should be published and mandated for use across all major national surveys to ensure consistency and comparability of data.

Proper application of survey weights:

The NSSO needs to ensure that the weighting methodology used in its surveys is robust and utilizes the most current population standards available, even in the absence of a recent Census. This might involve adjusting weights based on reliable population projections that take into account the latest trends in fertility, mortality, and migration. Furthermore, conducting sensitivity analyses

on the impact of different weighting assumptions on the final population estimates would help in understanding the potential range of variability.

Use of GIS and remote sensing technology:

The application of Geographic Information Systems (GIS) and remote sensing technology should be expanded for improved demarcation of administrative boundaries at all levels, including villages, hamlets, and urban areas. Remote sensing can be particularly valuable in identifying new settlements and monitoring changes in land use patterns, which can then be used to update sampling frames more accurately and efficiently.

Improved training for enumerators and survey staff:

Joint training programs for Census enumerators and NSS survey staff should be considered to ensure a consistent understanding of fundamental concepts, definitions, and enumeration procedures. These programs should emphasize the importance of thorough and accurate household listing, as well as the use of effective analytical techniques to ensure comprehensive data collection.

Incorporation of real-time data validation methods:

Both the Census and the NSS should increasingly incorporate digital data collection tools that include real-time data validation checks. This can help to minimize errors during the data entry process and improve the overall quality of the collected information.

Strengthening non-response follow-up mechanisms:

Standardized protocols for non-response follow-up should be developed and strictly implemented in both the Census and the NSS to mitigate the potential for bias introduced by non-participation. Additionally, strategies to improve response rates among groups that are typically reluctant to participate, such as

addressing concerns about data privacy and the burden of the survey, should be explored.

Developing a unified framework for population data integration:

A high-level national committee should be constituted to explore the feasibility of developing a unified framework for population data collection and estimation in India. This could involve examining the potential for a continuous population register or a more deeply integrated approach between the Census and the NSS, aiming to leverage the strengths of both methodologies to produce more consistent and timely population data.

CONCLUSION

The divergence in population estimates between the Census of India and the NSS survey arises from a complex relationship of methodological differences, practical challenges in data collection, and temporal mismatches in the availability of benchmark data. The Census gives a full count but becomes outdated over time, while the NSS provides frequent updates with some sampling limitations. Factors such as hamlet formation, outdated urban frames, varying reference periods, coverage issues, definitional problems, estimation techniques, sample size constraints, listing variability, and differing inclusion/exclusion criteria all contribute to the observed discrepancies. The delay in the Census 2021 has further compounded these issues.

Addressing this divergence is crucial as accurate population estimates are fundamental for informed policy decisions and effective resource allocation. Discrepancies can lead to inequities in welfare program coverage and delay evidence-based planning across various sectors. Bridging the gap between census and survey data is challenging, but aligning methods, enhancing data quality, and updating sampling frames are essential. The recommendations proposed in this report, focusing on regular frame updates, definition harmonization, proper weighting, technology utilization, enhanced training, real-time validation, strengthened non-response follow-up, and exploring a unified data framework, offer pathways towards achieving greater alignment in India's vital population statistics.

Ultimately, bridging these gaps is not just a statistical exercise, but a necessary step toward ensuring that national data systems serve the real and evolving needs of governance, development, and social equity. With deliberate reforms and sustained institutional commitment, the reliability and coherence of India's population statistics can be significantly enhanced.

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